

Project 2 - Autonomous, Onboard Vision-Based Trash and Litter Detection in Low Altitude Aerial Images

Introduction:

The project is an instance segmentation task where the individual pieces of trash in the images are identified, segmented and classified. The performance metric will be **mean Average Precision** over **IOU** of **0.5** and over **IOU** of **0.5-0.95**.

Approach

Model choice: Mask R-CNN, Yolo v8 and Yolo v8 + SAM were considered. Mask R-CNN while considered more accurate even though slower, Yolo v8 using the Ultralytics library was chosen as it was simpler to implement and also considered faster, considering the timeframe for the project..

Model architecture choice: Yolo v8 from Ultralytics offers nano(3.4 million params), small(11.8 mio), medium(27.3 mio), large(46 mio) and x-large models (71.8 mio). After running baseline models for 5 epochs, medium (mAP-0.20) was chosen over nano (mAP-0.12) and small (mAP-0.18). Large and X-large were not considered as they seem to overfit quickly..

Hardware used: Google Colab A100 GPU was used for training.

Data Preparation: An annotations json file was provided in "Coco" like format with 1500 annotated images. Of this, 62 images belonged to the test file and hence were used as test data for performance reporting purposes.

Baselining: Yolo v8 small was trained on 1438 images with 60% taken for training and the remaining for validation. The following were the results after 25 epochs. At this stage, 60 categories of trash were used for training.

Class	Images	Instances	Box(P)	R	mAP50	mAP50-95)	Mask(P)	R	mAP50	mAP50-95)
all	559	1883	0.358	0.238	0.197	0.162	0.355	0.236	0.192	0.148

mAP50 of 0.192 with a precision of 0.355 was achieved. On the baseline model. However, this does not generalise well on the test set. Hence data augmentation was performed.

Data Augmentation: Albumentations library was used to essentially perform the following transforms on the training images. These transforms were applied randomly in order to increase the diversity of the dataset..

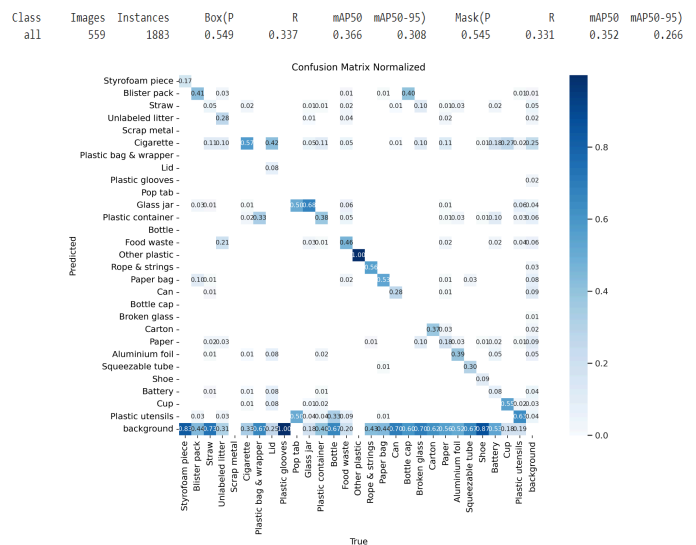
- Colour jitter
- Gaussian Noise
- Gaussian Blur
- Random Resized Crop
- Random Rotation

Up to 3 random transforms were applied on each training image, thus generating about 3200 images for training purposes. This vastly improved training results and also the results on the test set, which will be given below.

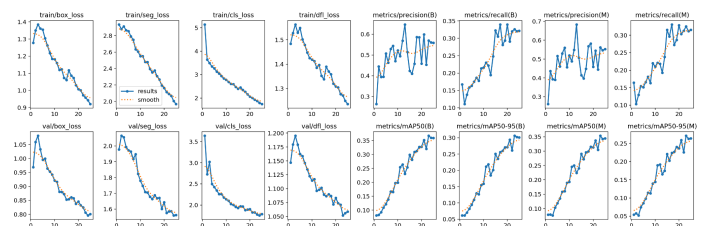
Category broadening: Instead of training for identifying 60 categories of trash, the model was trained to identify 28 super categories of trash as given in the annotations. This helped to improve generalisation on the test set.

Training: The training was run in 2 phases of 25 epochs each, with the second using the best parameters from the previous run. Here is the summary of the results of the two runs on the validation set of about 570 images. Default parameters of Yolo v8 were used.

First 25 epochs:



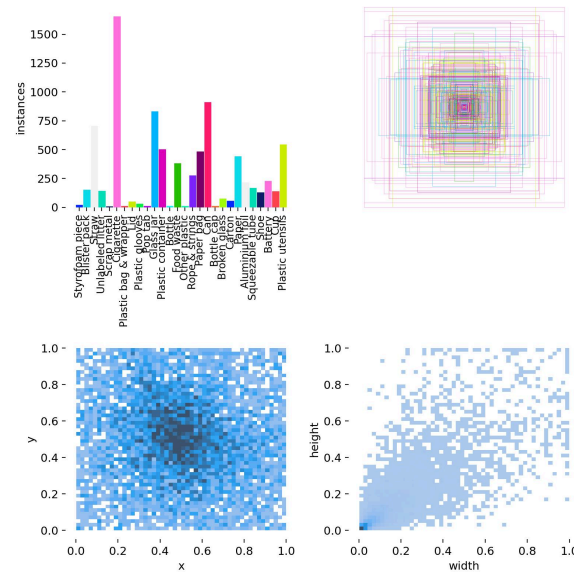
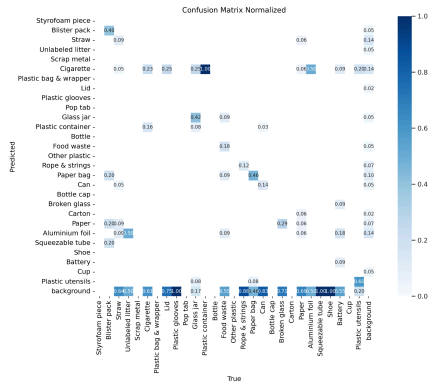
We can observe that the mAP50 has vastly improved to 0.352 over the baseline model. The Confusion matrix also has started to classify items better. We can also observe a rapid fall in losses and increase in precision, recall and mAP numbers.



The performance on the test set is:

Class	Images	Instances	Box(P)	R	mAP50	mAP50-95)	Mask(P)	R	mAP50	mAP50-95)
all	62	216	0.306	0.288	0.217	0.181	0.297	0.283	0.208	0.175

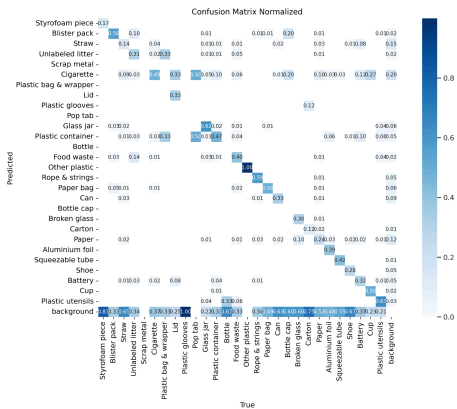
On the test set, the model gives a mAP50 of 0.208 for Mask segmentation. Visual verification of classification shows a pretty good detection of objects and segmentation, however, classification is short of ideal. Following is the confusion matrix on the test set.



Second 25 epochs: Another set of 25 epochs were run taking the best parameters of the first run. The results of the same on the validation set is presented below.

Class	Images	Instances	Box(P)	R	mAP50	mAP50-95	Mask(P)	R	mAP50	mAP50-95
all	559	1883	0.549	0.365	0.38	0.322	0.539	0.345	0.363	0.281

The mAP50 for the mask has improved to 0.363 while precision and recall had marginally improved. The confusion matrix on the validation set is as below.



Class	Images	Instances	Box(P)	R	mAP50	mAP50-95	Mask(P)	R	mAP50	mAP50-95
all	559	1883	0.549	0.337	0.366	0.308	0.545	0.331	0.352	0.266
Styrofoam piece	5	6	1	0.232	0.559	0.503	1	0.232	0.559	0.326
Blister pack	33	39	0.585	0.487	0.45	0.383	0.585	0.487	0.446	0.321
Unlabeled litter	115	200	0.29	0.115	0.13	0.072	0.278	0.11	0.116	0.0572
Cigarette	26	29	0.411	0.448	0.388	0.368	0.411	0.448	0.387	0.35
lastic bag & wrapper	189	274	0.409	0.602	0.471	0.367	0.404	0.595	0.46	0.341
Lid	3	3	1	0	0.337	0.337	1	0	0.337	0.337
Bottle	92	101	0.516	0.167	0.324	0.286	0.51	0.167	0.324	0.316
Plastic gloves	3	5	0	0	0	0	0	0	0	0
Pop tab	2	2	1	0	0.06	0.049	1	0	0.06	0.0419
Glass jar	132	185	0.736	0.697	0.776	0.658	0.742	0.703	0.778	0.607
Plastic container	70	100	0.471	0.5	0.417	0.363	0.452	0.48	0.383	0.293
Bottle	2	3	1	0	0.0356	0.0356	1	0	0.0356	0.0333
Food waste	75	82	0.482	0.549	0.54	0.449	0.461	0.524	0.517	0.4
Other plastic	1	1	0.498	1	0.995	0.995	0.498	1	0.995	0.895
Rope & strings	42	70	0.685	0.586	0.616	0.436	0.699	0.598	0.636	0.304
Paper bag	93	101	0.516	0.569	0.504	0.395	0.507	0.559	0.481	0.313
Can	93	321	0.596	0.285	0.38	0.22	0.583	0.278	0.325	0.147
Bottle cap	4	5	1	0	0.00393	0.00354	1	0	0.00393	0.00393
Broken glass	7	10	0.164	0.1	0.0573	0.0498	0.164	0.1	0.0573	0.0483
Carton	7	8	0.176	0.375	0.347	0.251	0.117	0.25	0.123	0.0985
Paper	58	87	0.178	0.184	0.175	0.145	0.119	0.195	0.179	0.127
Aluminium foil	22	31	0.255	0.419	0.348	0.339	0.255	0.419	0.348	0.32
Squeezable tube	27	33	0.739	0.303	0.388	0.303	0.739	0.303	0.376	0.243
Shoe	8	86	0.925	0.105	0.241	0.136	0.925	0.105	0.234	0.0958
Battery	45	60	0.27	0.167	0.169	0.14	0.27	0.167	0.169	0.144
Cup	17	22	0.338	0.545	0.528	0.496	0.338	0.545	0.528	0.492
Plastic utensils	71	108	0.594	0.667	0.652	0.546	0.594	0.667	0.651	0.527

Visual examples of prediction:

Here are some examples of prediction on the test image sets.

Ground Truth 1	Prediction 1
Ground Truth 2	Prediction 2
Ground Truth 3	Prediction 3

However, the results of the model on the test set were not better than the model at the end of the first 25 epochs, suggesting that the model was overfitting. The results are shown below.

Class	Images	Instances	Box(P)	R	mAP50	mAP50-95	Mask(P)	R	mAP50	mAP50-95
all	62	216	0.243	0.302	0.209	0.18	0.239	0.293	0.206	0.168

The mAP50 for masks had marginally fallen down to 0.206. The precision and recall numbers have also fallen indicating overfitting. Therefore, the parameters at the end of the first run of 25 epochs is considered as the best at the end of this exercise. The confusion matrix had also become distorted at the end of the second run.

Category wise observations:

With the best parameters from the first run of 25 epochs, the results on the validation set is given in the table below. One can observe that categories like Styrofoam piece, Cigarette, Glass Jar, Plastic Container, Cup and Plastic utensils have a fairly high mAP50, Precision and Recall. Consequently, those items are more easily detected and segmented. Cigarette classes are overwhelmingly high in the training data set followed by can. Similarly, items of very small size also dominate the dataset.